

OPTIMIZATION AND SIMULATION

FINAL PROJECT

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Juan Díaz Arenas

Javier Andrés Bernárdez

Juan García Moraga

Ana Banqueri Camy

1. Introduction

Wednesday, June 25th. 9:00h

The Philadelphia 76ers front office enters the team's headquarters as if it were a normal day. The 24-25 NBA season finished just three days ago, with the Oklahoma City Thunder crowning themselves as "World Champions", and the preparation for the next one starts right now.

But today is no normal day. The NBA draft is being held tonight in Brooklyn, and the 76ers have the third pick after a disappointing season where they missed the playoffs. However, with a good young player joining the team and a healthy Joel Embiid coming back, the 76ers could finally win the Eastern Conference, especially with both Pacers and Celtics, the last two East champions, losing star players to injury.

Monday, June 30th. 19:00h

With VJ Edgecombe selected and free agency a few hours away, the 76ers need to finalize their roster construction in order to compete for next season's title. Nevertheless, there are some internal concerns regarding Joel Embiid's health issues, so the general manager has secretly hired four mathematical engineers, experts in modeling, disguised as summer interns to analyze the viability of Embiid being the centerpiece of the team...

2. The process

Tuesday, July 1st. 10:00h

First day in the building for the new signings. After a quick introduction to the rest of the team, the general manager brings them to their office, where he explains what their roles will consist of. He tries to keep it simple: they need to help the front office predict a realistic probability of Embiid getting hurt before the playoffs in order to make a decision before the season starts.

The general manager states several additional conditions. Although important to manage Embiid's workload, the 76ers must make the playoffs, preferably avoiding the play-in. Looking at every season since this mini-tournament was implemented, the average number of wins for the **6 seed** in the Eastern Conference is **45**. The NBA obliges star players to suit up for In-Season Tournament games, which means that Embiid will play no less than the four scheduled group stage games.

Embiid also has some personal demands. As the star player he is, he wants to take part in at least **65 games** to be eligible for regular season awards and All-NBA teams. If he were to play less than **1000 minutes** through the entire season, he would ask for a trade to another team. Finally, he is scared to play against the Nikola Jokic led Nuggets on the road, which he hasn't done since November 8th, 2019, and, to be coherent with our simulation exercise, he can't suit up on Friday, November 28th (@ Brooklyn Nets), as he will be spending the afternoon with his family in a shopping mall. Despite the latter being an In-Season Tournament game, the franchise will happily take the fine in exchange for Embiid being happy. Work starts. The 76ers have no time to lose.

Tuesday, July 1st. 18:00h

The first day of work has finished, but, before leaving, the hired team has a meeting with the general manager. They have already come up with a model to predict Embiid's health throughout the season. The plan is simple: his fitness starts at a maximum at the beginning of the season and goes down with every game he plays, slightly recovering after each missed fixture. The general manager then suggests that Embiid's fitness at the end of the regular season should be no lower than 75%, as a lower number would jeopardize the team's chances of a potential playoff run. The formula that the modeling team has come up is the following one:

$$F_g = F_{g-1} - 0.00025 \cdot X_{g-1} - 0.000125 \cdot X_{g-2} - 0.0000625 \cdot X_{g-3} + 0.012 \cdot (1 - Y_{g-1})$$
$$\forall g \in G$$

In order to simplify the vast amount of games played and the staggering twenty-nine opponents faced, the mathematical engineers have decided to divide the NBA franchises into four different clusters: elite, good, average and bad. This classification is made based on every team's net rating (point differential for every hundred possessions) in the 24-25

season and the number of wins projected for the upcoming season. The clustering results are the following:

- **Elite:** Cleveland Cavaliers, Denver Nuggets, Houston Rockets, New York Knicks, Oklahoma City Thunder (14 games played)
- **Good:** Atlanta Hawks, Boston Celtics, Detroit Pistons, Golden State Warriors, Los Angeles Clippers, Los Angeles Lakers, Milwaukee Bucks, Minnesota Timberwolves, Orlando Magic, San Antonio Spurs (28 games played)
- **Average:** Chicago Bulls, Dallas Mavericks, Indiana Pacers, Memphis Grizzlies, Miami Heat, Phoenix Suns, Portland Trail Blazers, Sacramento Kings, Toronto Raptors (25 games played)
- **Bad:** Brooklyn Nets, Charlotte Hornets, New Orleans Pelicans, Utah Jazz, Washington Wizards (15 games played)

Sidenote: As this project isn't being done in July 2025, we, as of January 2026, have information about the teams' performances during this season. When comparing predicted clusters to reality, we can observe how more or less they correspond with what's happening in the NBA as of today.

Once the opponents are classified into four different clusters, it is time to calculate the win probability of the 76ers against them. To do this, the team decides to use the Normal CDF, with which they can estimate a final probability based on the comparison of both teams net rating and the number of game possessions, with the 76ers average during the past five seasons being **98.17**. In addition, a standard deviation of **12** possessions is assumed for the model, as there are teams that play at higher and lower paces. The final formula is the next one:

$$F(x) = \frac{1}{2} \left[1 + \operatorname{erf} \left(\frac{x - \mu}{\sigma \sqrt{2}} \right) \right]$$

Now they identify the 76ers net rating in different situations (data through the last five seasons):

- Embiid plays at home: 6.4
- Embiid plays on the road: 4.2
- Embiid doesn't play at home: -3.7
- Embiid doesn't play on the road: -3.8

After estimating every parameter, the mathematical engineers present their win probability against each cluster, depending on Embiid playing or not and the game being at home or on the road:

Elite teams:

- At home with Embiid: 0.48

- At home without Embiid: 0.19
- On the road with Embiid: 0.41
- On the road without Embiid: 0.19

Good teams:

- At home with Embiid: 0.63
- At home without Embiid: 0.31
- On the road with Embiid: 0.56
- On the road without Embiid: 0.3

Average teams:

- At home with Embiid: 0.72
- At home without Embiid: 0.4
- On the road with Embiid: 0.65
- On the road without Embiid: 0.4

Bad teams:

- At home with Embiid: 0.9
- At home without Embiid: 0.68
- On the road with Embiid: 0.87
- On the road without Embiid: 0.68

The team notices that the 76ers are an almost contender when Embiid is on the floor vs a below average team when he doesn't play, which more or less fits the past few seasons. They have also discovered that Embiid playing less than **24 minutes** is equivalent to him not playing and that his contribution to the team caps at **38 minutes**, with every value in between corresponding to a linear interpolation between the win probability with and without Embiid.

Finally, the team's goal is to minimize the deviation from the number of games that Embiid wants to play and the one from the fitness that the franchise expects, in addition to not wanting a trade demand by Embiid.

The workday ends and everyone returns to their homes. The general manager takes a last look at the paper with the complete model of the situation before closing his office's door.

SETS

game = {1, 2, ..., 82}
ratings = {elite, good, average, bad}
place = {home, away}

PARAMETERS

P_{gr} : Game g played against rating r

I_{gp} : Game g played at place p

W_{rp} : Winning % against team of rating r at place p without Embiid

S_{rp} : Winning % added when Embiid plays against rating r at place p

T_g : In-Season tournament game in game g

VARIABLES

X_g : Embiid's minutes in game g

Y_g : Embiid plays game g

Z : Embiid demands trade

F_g : Embiid's fitness in game g

OBJECTIVE

$$\min(W_n \cdot n + W_f \cdot f + W_z \cdot Z)$$

Where:

- n : Deviation of games played $n = \max(0, 65 - \sum Y_g)$
- f : Deviation of fitness lower than 75% $f = \sum_g \max(0, 0.75 - F_g)$
- Z : Penalty if Embiid asks for a trade

Parameter weights:

- W_n : Importance of reaching 65 games played (MVP eligibility)
- W_f : Importance of entering the playoffs fit
- W_z : Importance of avoiding a trade demand

The constraints associated to the slack variables are the following:

- **Games played slack (n):** $n \geq 65 - \sum Y_g$.
- **Fitness slack (f):** $f \geq 0.75 - F_{82}$.

CONSTRAINTS

- How fresh will be Embiid on any game:

$$F_g = F_{g-1} - 0.00025 \cdot X_{g-1} - 0.000125 \cdot X_{g-2} - 0.0000625 \cdot X_{g-3} + 0.012 \cdot (1 - Y_{g-1})$$
$$\forall g \in G$$

- Limit of time played: $24 \cdot Y_g \leq X_g \leq 38 \cdot Y_g \quad \forall g \in G$

- Embiid will not play the match on Black Friday (28 November): $Y_{18} = 0$
- Embiid will not play at Denver: $Y_{69} = 0$
- Minimum number of victories required:

$$\sum_{g \in G} \sum_{r \in R} \sum_{p \in P} I_{g,p} \cdot P_{g,r} \cdot \left(W_{rp} + \frac{S_{rp}}{0.58} \cdot \left(\frac{X_g}{24} - Y_g \right) \right) \geq 45$$

- Embiid must play matches marked as tournament: $Y_g \geq T_g$
- Embiid must play a minimum of 1000 minutes in total by the end of the season, otherwise the trade restriction Z will activate to fulfill the constraint.

$$\sum_g X_g + 5000 \cdot Z \geq 1000$$

3. The results

Wednesday, July 2nd. 9:00h

The team arrives at the general manager's office to run yesterday's model.

$$\min(5 \cdot n + 20 \cdot f + 1000 \cdot Z)$$

The results obtained are the following:

	Indicador	Valor	Estado
0	Función Objetivo (Minimizar)	9.6532	Optimizado
1	Partidos Jugados (Total)	65.0000	65
2	Partidos Faltantes para 65 (n)	0.0000	OK
3	Estado Físico Final Real (F_82)	0.2673	26.73%
4	Objetivo Físico Deseado	0.7500	75.00%
5	Déficit Forma Física (f)	0.4827	Déficit
6	Pide Traspaso (Z)	0.0000	NO
7	Victorias Esperadas Totales (>45)	45.0000	Cumple

The general manager doesn't like what he sees. He thinks the team has given too much importance to Embiid's 65 game demand, which results in him entering the playoffs at 26.73% fitness, a number far from the 75% the general manager initially asked for. He rapidly kicks the team out of his office, urging them to prepare another model that adjusts better to the 76ers' demand before the end of the day.

Wednesday, July 2nd. 19:00h

The team has spent the whole day working on the new model, and it is now time to present it to the general manager. The latter is far more relieved after analysing the results, as, although Embiid is not close to qualifying for individual awards with only 49 games played, he enters the playoffs at around 70% fitness, which is a much better number.

	Indicador	Valor	Estado
0	Función Objetivo (Minimizar)	24.5261	Optimizado
1	Partidos Jugados (Total)	46.0000	46
2	Partidos Faltantes para 65 (n)	19.0000	Falta
3	Estado Físico Final Real (F_82)	0.6947	69.47%
4	Objetivo Físico Deseado	0.7500	75.00%
5	Déficit Forma Física (f)	0.0553	Déficit
6	Pide Traspaso (Z)	0.0000	NO
7	Victorias Esperadas Totales (>45)	45.0000	Cumple

The new objective function is the following:

$$\min(1 \cdot n + 100 \cdot f + 1000 \cdot Z)$$

Although the fitness with which Embiid enters the playoffs is a bit smaller than the general manager was expecting, he is forced to accept it, as the 76ers barely reach 45 wins, which is absolutely mandatory for the season to not be considered a failure.

The general manager is thus less concerned about the possibility of Embiid forcing his way out of Philadelphia, as if the 76ers want to make the playoffs he needs to play a higher amount of minutes than the set threshold, so he asks the team for a last time to analyze the model and see if, in practically the worst case scenario, Embiid's minutes are close to being below 1000.

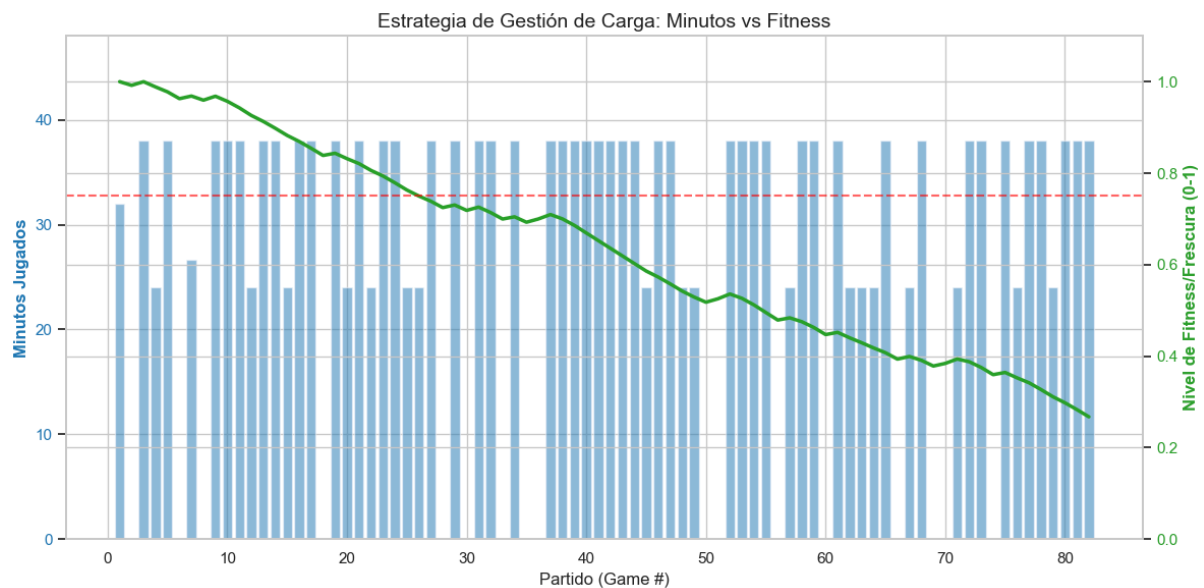
The team comes back a while later and the general manager is suddenly relieved. Even in the worst case the team predicted that Embiid should play around 1750 minutes, which is very far away from the previous threshold of 1000 and more than enough to feel comfortable about Embiid never requesting a trade due to lack of minutes.

Minutos Totales Jugados	1744.882522238266	1744
Promedio Minutos (Temporada Com	21.279055149247146	21.3 m/p

4. The analysis

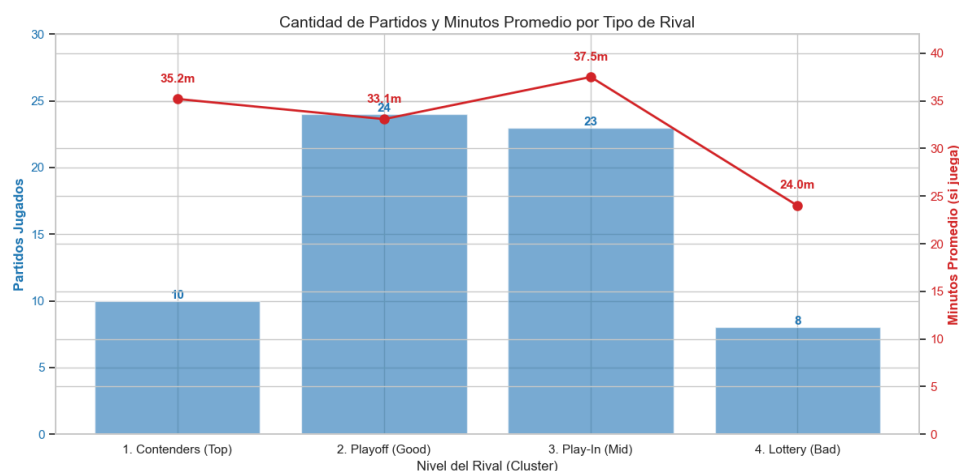
Thursday, July 3rd. 9:00h

New day at the office and it is now time to analyze the results. The team has prepared several graphs to see the differences between models. Despite the general manager's refusals, they start with the one that prioritizes Embiid's personal demands.



After being below 75% for the first time around the 25th game, Embiid doesn't recover and ends up in a catastrophic value of 26.73%. Although there are several games in which he doesn't play, it is still not enough to compensate for the huge amount in which he does.

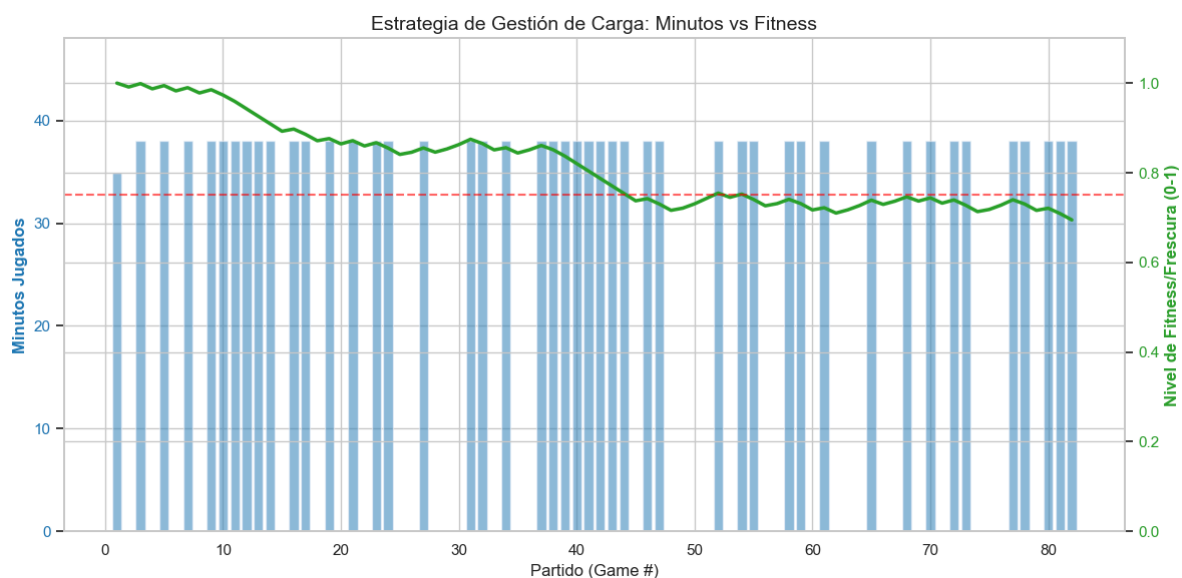
Analyzing Embiid's played games distribution depending on the opponent is also valuable for the team, so their next graph is the following:



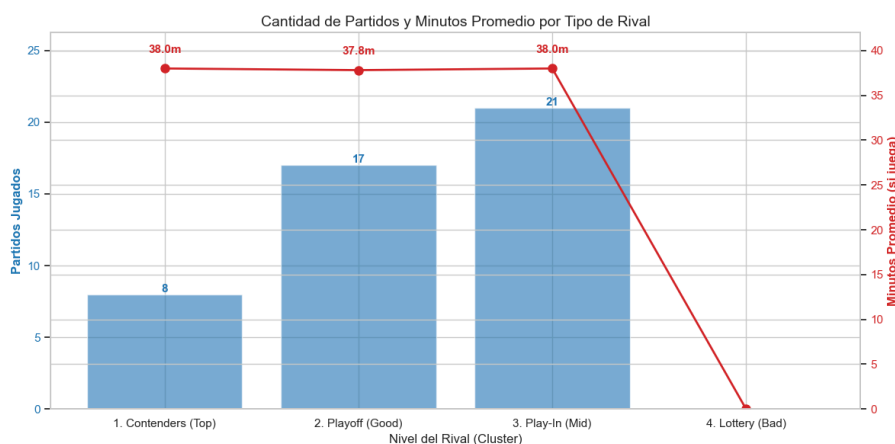
Which shows that the models estimates that it is more optimal for Embiid to play more minutes against clusters with higher differences in win probability when he does and doesn't play, which are mainly the first three ones.

The team finds it interesting that he averages 24 minutes against bad teams, which results in him not adding any win probability and downgrading his fitness in exchange for being closer to 65 games. At this point of the presentation, the general manager forces them to move on to the next model, as he doesn't want to see more of what he considers "selfish".

The team now brings up the graphs for the model that gives more importance to Embiid being fit at the end of the season. As with the previous case, they start off by showing the progression of fitness throughout the season.



In this model, Embiid now plays less games with fewer minutes, leading to better fitness at the end of the season. However, this isn't enough to satisfy the general manager's initial demands. This is reflected on the next graph:



Where Embiid again plays almost the full 38 minutes that contribute to the team winning against the three best clusters, but this time, he doesn't suit up for meaningless games, playing a total of zero games against bad teams. The general manager radically prefers this approach, and congratulates the team for their efforts.

As changing the constraint about Embiid asking for a trade due to lack of minutes does not have an impact on the results, the team considered that it wasn't necessary to analyze a model where a trade is less penalized. Furthermore, the general manager doesn't want to hear "Embiid" and "trade" in the same sentence.

5. The future

Friday, August 1st. 9:00h

After completing their jobs, yesterday was the last day at the office for the mathematical engineers. The 76ers's GM enters his office as if it were a normal day. Today is, indeed, a normal day. He lays in his seat, breathing a sight of relief, thinking the Embiid situation is under control. Suddenly, the phone rings. It's the general manager of the Philadelphia Eagles, a good friend of his. He wants to know the contact information of the four mathematical engineers, as he is keen on solving AJ Brown's conflicts with the team. **To be continued...**

Credits: *Basketball Reference*, *Statmuse*

Link to Github: <https://github.com/juanmoraga05/ProyectoOptimizacion.git>